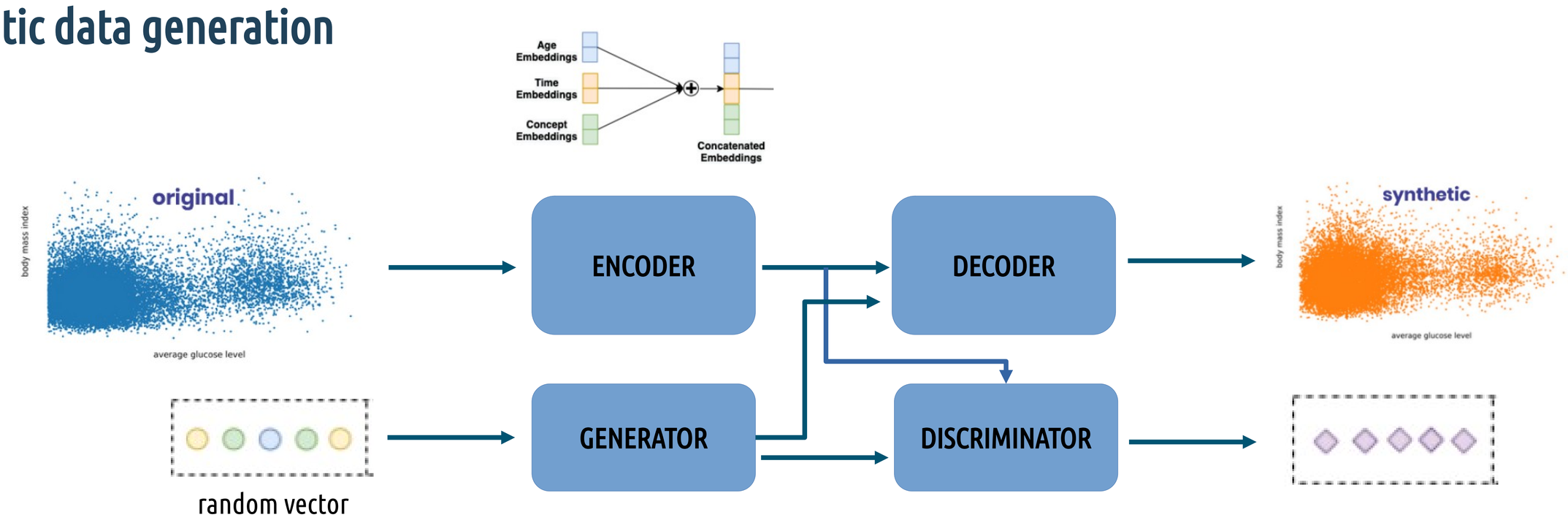
Generating Synthetic Data from OMOP-CDM Databases for Health Applications

A. Labarga[1,2] , S. Aguiló-Castillo[1,2], S. Capella-Gutierrez[1,2]
[1] Barcelona Supercomputing Center (BSC), Barcelona Spain.
[2] Spanish National Bioinformatics Institute (INB/ELIXIR-ES).

Analysis of Electronic Health Records (EHR) has a tremendous potential for enhancing patient care, quantitatively measuring performance of clinical practices, and facilitating clinical research. Statistical estimation and machine learning (ML) models trained on EHR data can be used to predict the probability of various diseases (such as diabetes), track patient wellness, and predict how patients respond to specific drugs. For such models, researchers and practitioners need access to EHR data. However, it can be challenging to leverage EHR data while ensuring data privacy and conforming to patient confidentiality regulations. Here we present an approach fo generating synthetic health data from an OMOP-CDM. The goal of this study was to develop and evaluate a model for simulating longitudinal healthcare data that adequately captures clinical data temporal and conditional complexities.

Synthetic data generation

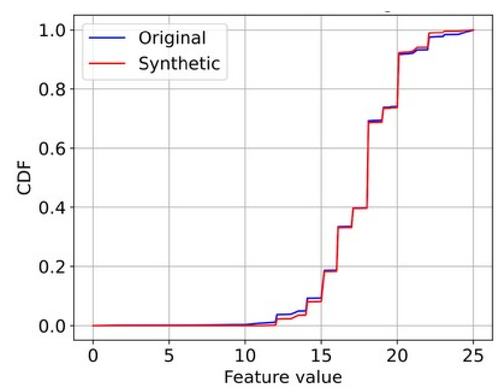


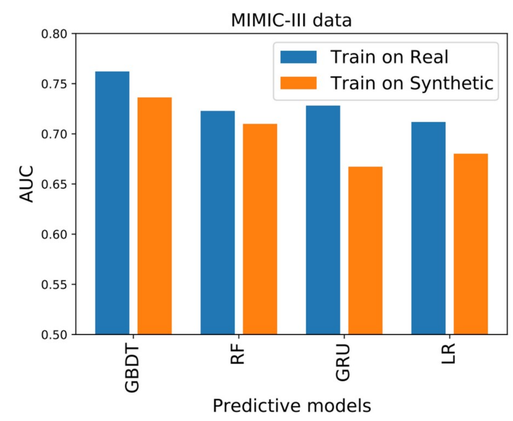
Generating synthetic data comes down to learning the joint probability distribution in an original, real dataset to generate a new dataset with the same distribution. Deep learning models such as **generative adversarial networks (GAN)** and **variational autoencoders (VAE)** are well suited for synthetic data generation but have problems capturing temporal and causal dependencies in the data or generating categorical variables common in clinical data.

We propose a novel generative modeling framework that combines GANs with a **bidirectional encoder representations from transformers (BERT)** architecture. We first train the encoder-decoder model using a reconstruction loss. Then, we use the trained encoder to transform the original inputs into latent space (encoder states). Lastly, we train the GAN framework using an adversarial loss in the latent space to incorporate temporal data across multiple clinical domains. We use a hybrid approach by augmenting the input to BERT using artificial time tokens, incorporating time, age, and concept embeddings, and introducing a new second learning objective for visit type.

Quality evaluation

FIDELITY	UTILITY	PRIVACY
How similar is this synthetic data as compared to the original training sets	How useful is this synthetic data for our downstream machine learning applications	Has any sensitive data been inadvertently synthesized by our model
Kullback-Leibler (KL) divergence, pairwise correlation difference	Accuracy, F1-score, ROC, and AUC-ROC	Membership inference, re-identification and attribute inference attacks





References

1. Murray Reet al. Design and validation of a data simulation model for longitudinal healthcare data. AMIA Annu Symp Proc. 2011
2. Pang etal. CEHR-BERT: Incorporating temporal information from structured EHR data to improve prediction tasks. Proc. of Machine Learning for Health, 158, 2021.
3. Yoon et al. EHR-Safe: Generating High-Fidelity and Privacy-Preserving Synthetic Electronic Health Records https://doi.org/10.21203/rs.3.rs-2347130/v1

Contact

Alberto Labarga
alberto.labarga@bsc.es

Barcelona Supercomputing Center (BSC)
Plaça Eusebi Güell, 1-3. 08034 Barcelona, Spain

Funding

This work has been funded by the Institute of Health Carlos III (project IMPaCT-Data, exp. IMP/00019), co-funded by the European Union, European Regional Development Fund (ERDF, "A way to make Europe") and from the European Union's Horizon Europe Programme under GA 101058427 — EOSC4Cancer — HORIZON-INFRA-2021-EOSC-01